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Time Series & Text Analysis

> Relevant source files

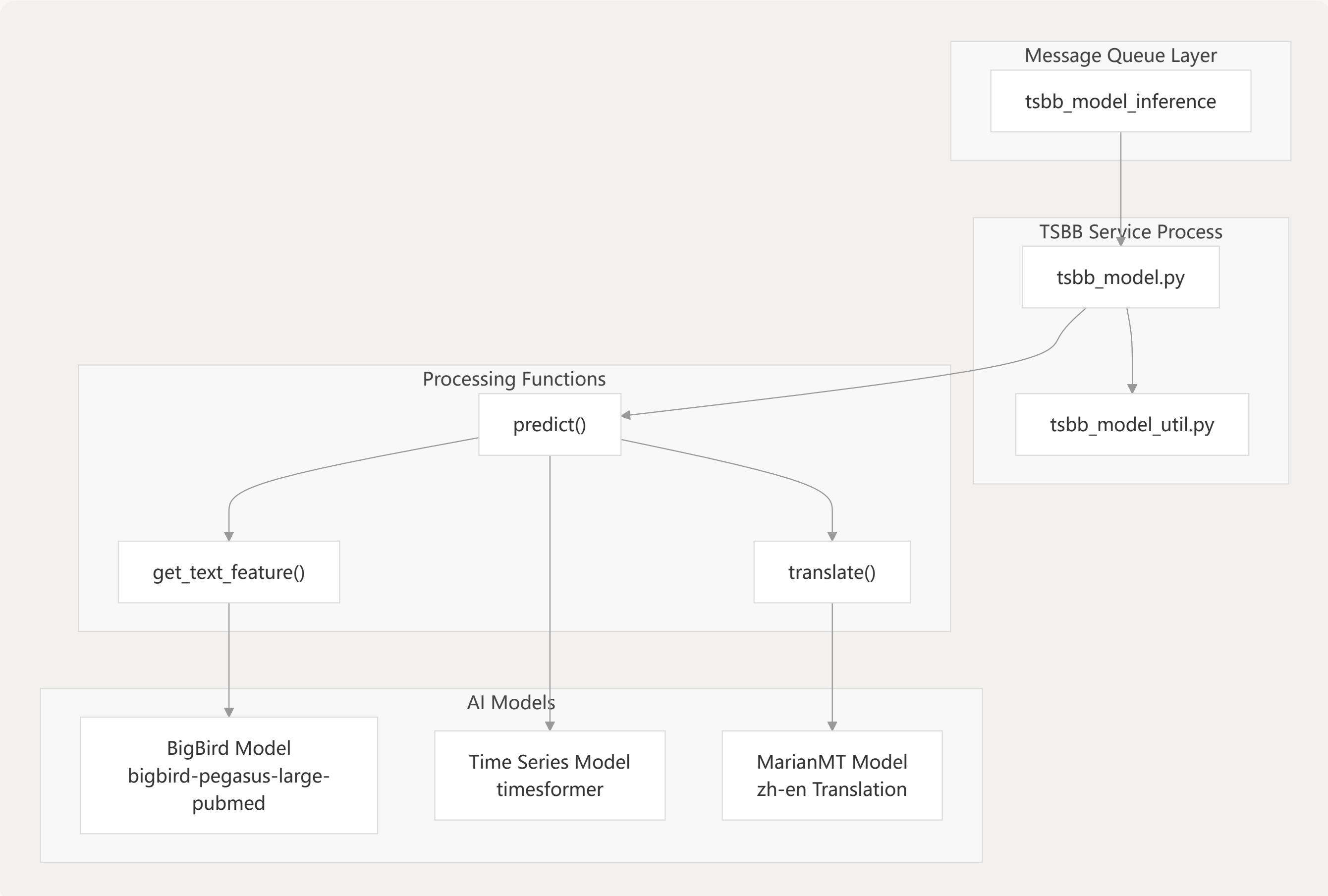
This document covers the Time Series & Text Analysis service module of the MKTY Smart Healthcare System, which provides advanced natural language processing and time series prediction capabilities for medical applications. The service combines BigBird transformer models for text feature extraction with time series prediction models to support medical text analysis and temporal data forecasting.

For large language model services, see Large Language Model Service. For multimodal AI services involving image analysis, see Multimodal AI Services.

Architecture Overview

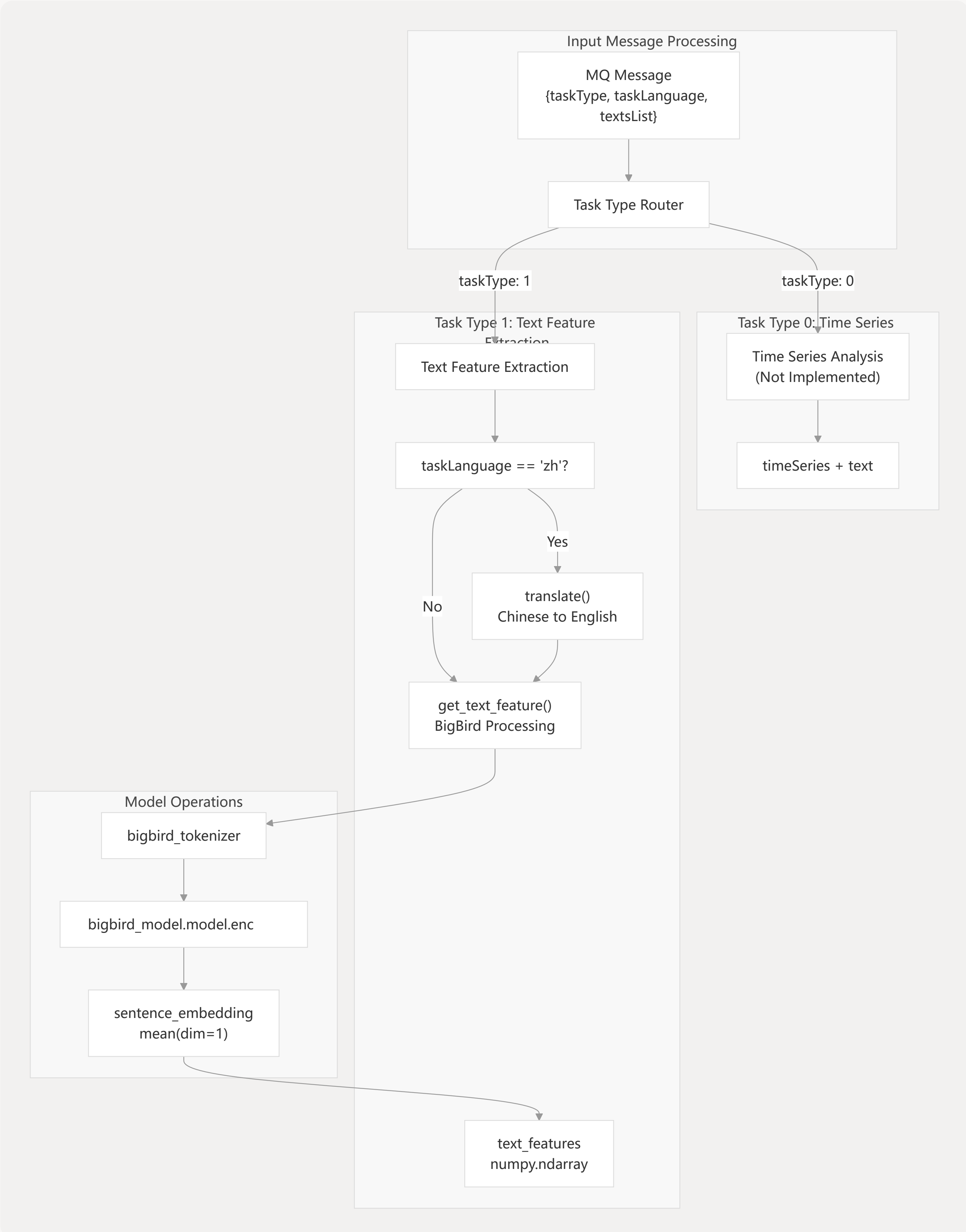
The TSBB (Time Series BigBird) service operates as a RabbitMQ message queue consumer that processes requests for text feature extraction and time series analysis. The service integrates multiple AI models to provide comprehensive text and temporal data analysis capabilities.

TSBB Service Architecture



Sources: `backend/tsbb_model.py` 1-84 `backend/tsbb_model_util.py` 1-152

Task Processing Workflow



Sources: `backend/tsbb_model.py` 24-67 `backend/tsbb_model.py` 35-52

Core Components

BigBird Text Feature Extraction

The service uses the BigBird Pegasus model specifically trained on PubMed data for medical text analysis. The `get_text_feature` function extracts semantic embeddings from input text.

Component	Implementation	Purpose
Model	`BigBirdPegasusForConditionalGeneration`	Medical text understanding
Tokenizer	`AutoTokenizer`	Text preprocessing
Feature Type	Sentence embeddings	Global text representation
Output Dimension	Variable (model-dependent)	Dense feature vectors

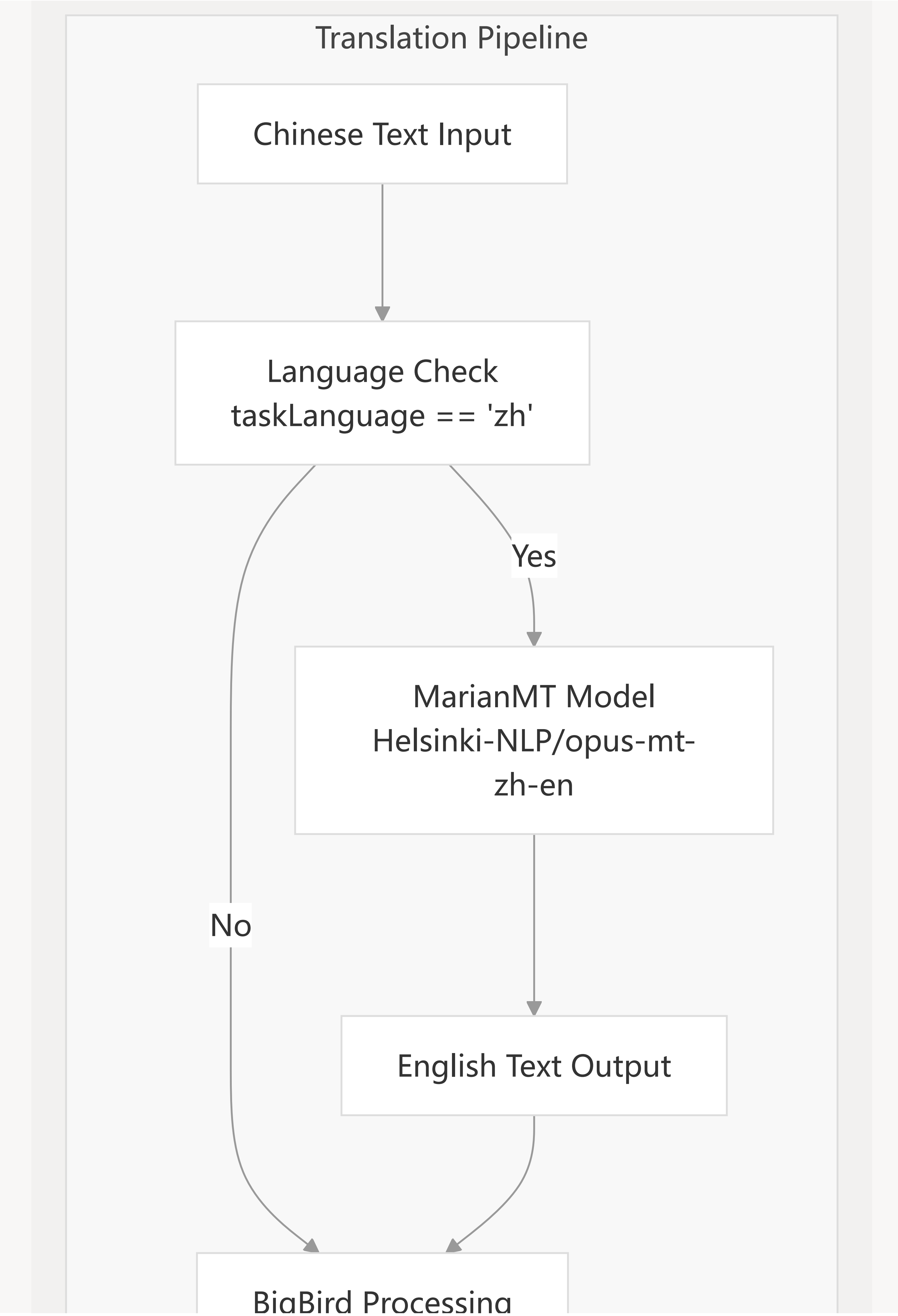
The text feature extraction process:

1. **Tokenization**: Input text is tokenized using `bigbird_tokenizer`
2. **Encoding**: Tokens are processed through `bigbird_model.model.encoder`
3. **Embedding Extraction**: Mean pooling across token embeddings to create sentence-level features
4. **Output**: Returns dense numpy arrays representing text semantics

Sources: `backend/tsbb_model.py` 35-52 `backend/tsbb_model_util.py` 96-111

Translation Integration

The service integrates MarianMT translation to support Chinese medical texts by translating them to English before processing with the English-trained BigBird model.



Translation Pipeline
Chinese Text Input
Language Check
taskLanguage == 'zh'
Yes
No
MarianMT Model
Helsinki-NLP/opus-mt-zh-en
English Text Output
BigBird Processing

Sources: `backend/tsbb_model.py` 63-64 `backend/tsbb_model_util.py` 117-151

Time Series Model Integration

The time series component is designed to work with temporal medical data, though the implementation is currently a placeholder in the codebase.

Component	Status	Configuration
Model Path	`./backend/timesformer`	Directory for time series models
Loader Function	`load_model_and_tokenizer_ts()`	Placeholder implementation
Task Type	0	Combined time series + text analysis
Implementation	Not complete	Future development

Sources: `backend/tsbb_model.py` 19 `backend/tsbb_model_util.py` 113-115

Message Queue Integration

Queue Configuration

The TSBB service operates as a RabbitMQ consumer with the following configuration:

```
MQ_CONNECTION_PARAMETERS = {
    'host': 'localhost',
    'port': 5672,
    'heartbeat': 0
}
MQ_NAME = 'tsbb_model_inference'
```

Message Format

The service expects JSON messages with the following structure:

Field	Type	Description	Required
`taskType`	int	Task type (0: time series, 1: text)	Yes

Field	Type	Description	Required
`taskLanguage`	string	Language code ("zh", "en")	Yes
`textsList`	array	List of text strings to process	For taskType 1
`timeSeries`	array	Time series data points	For taskType 0
`text`	string	Additional text context	For taskType 0

Response Format

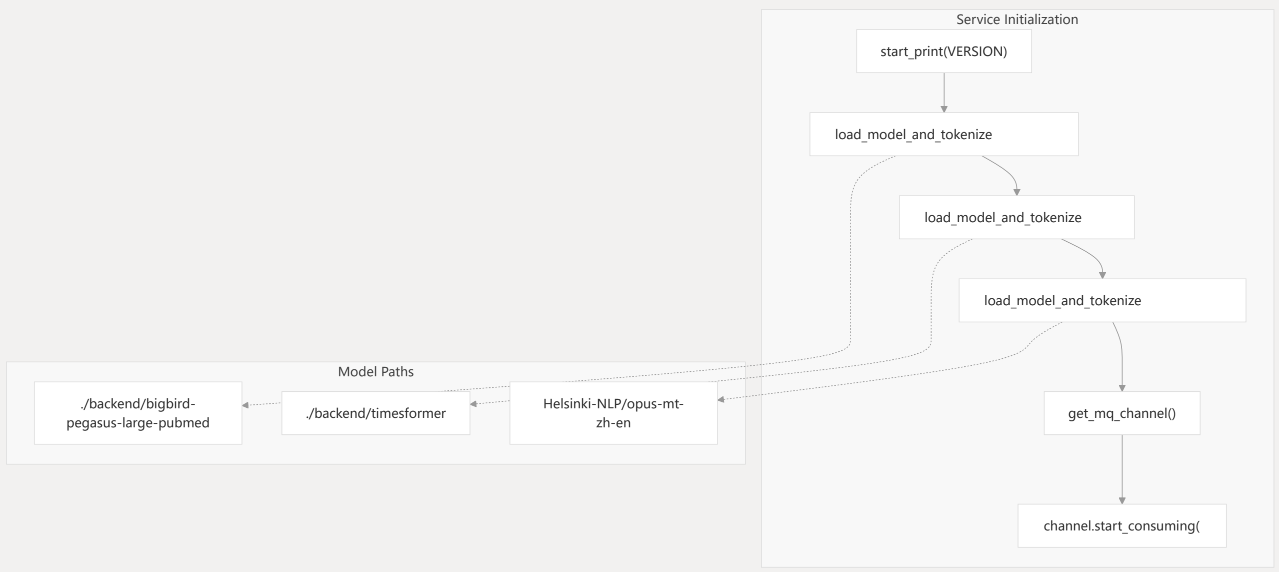
The service returns structured responses based on task type:

```
{
  "taskType": 1,
  "data": [[0.1, 0.2, ...], [0.3, 0.4, ...]]
}
```

Sources: `backend/tsbb_model.py` 11-16 `backend/tsbb_model.py` 53-67

Model Loading and Initialization

Startup Sequence



Sources: `backend/tsbb_model.py` 22-83

Model Parameters